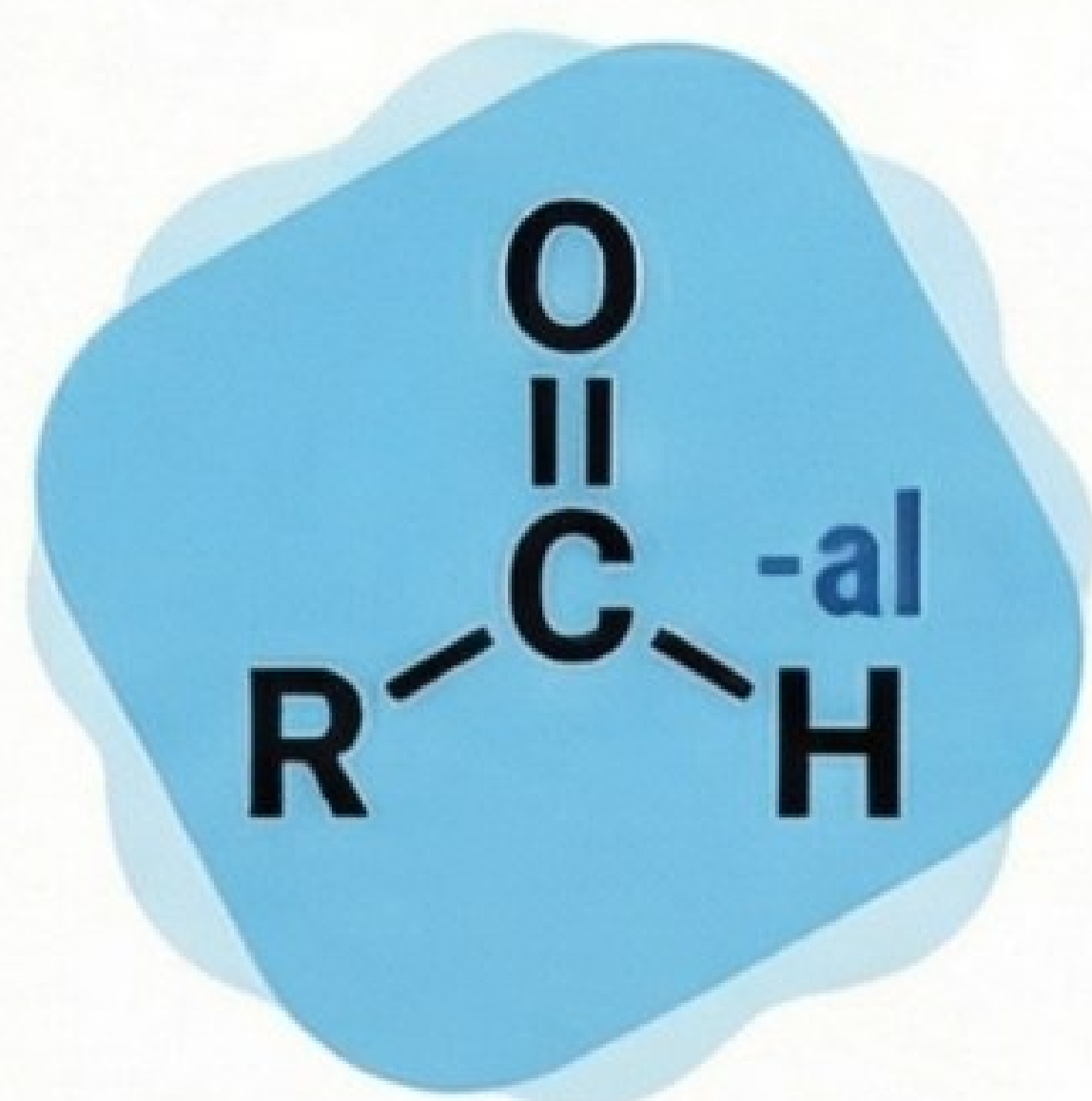


A Visual Guide to Aldehydes, Ketones & Carboxylic Acids

IDENTIFYING THE KEY PLAYERS

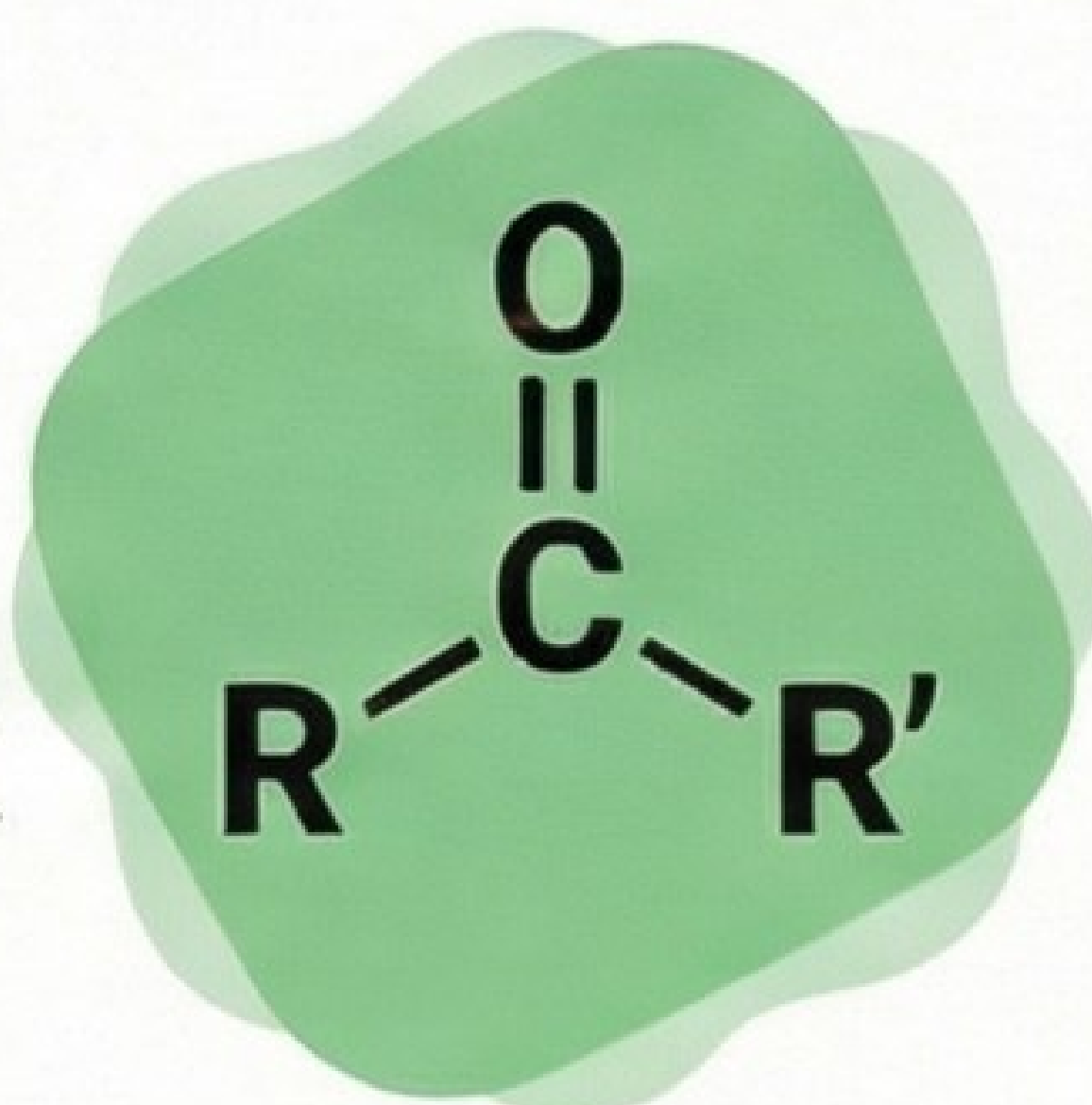


ALDEHYDE R-CHO

Characterized by a carbonyl group (C=O) bonded to at least one hydrogen atom.

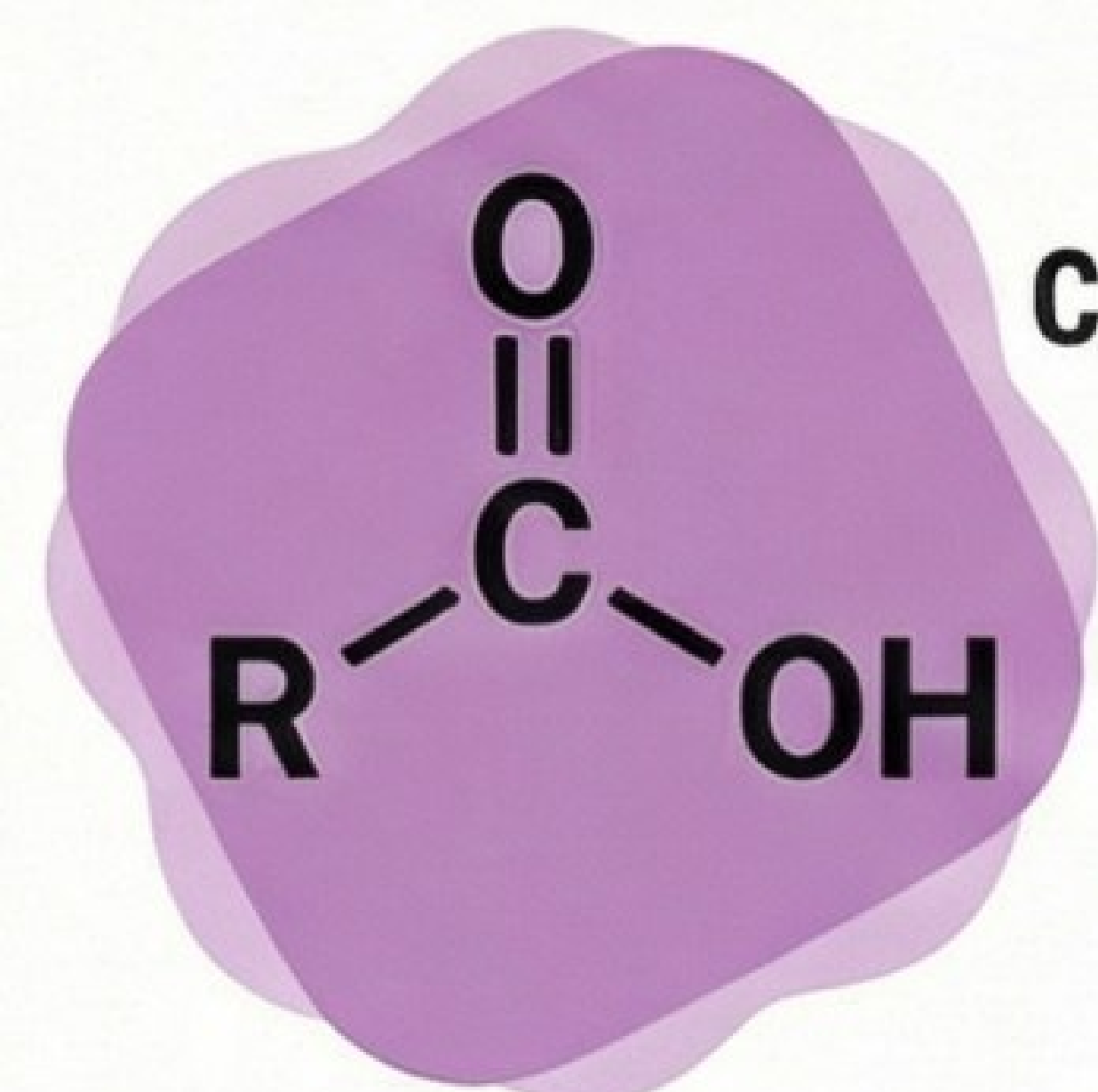
KETONE -one

Features a carbonyl group (C=O) bonded to two carbon atoms.



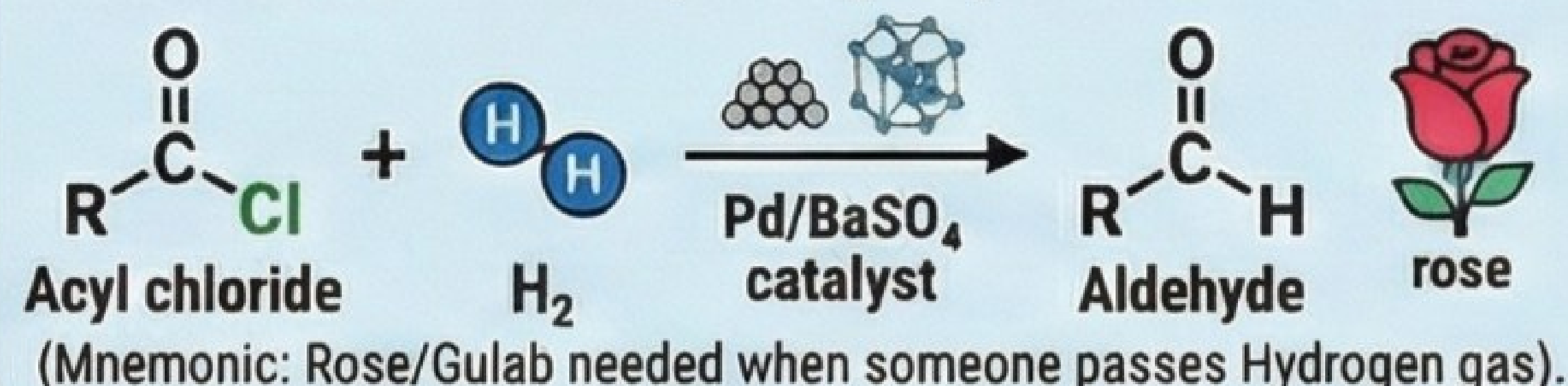
CARBOXYLIC ACID -oic acid

Contains a carbonyl group (C=O) attached to a hydroxyl group (-OH).

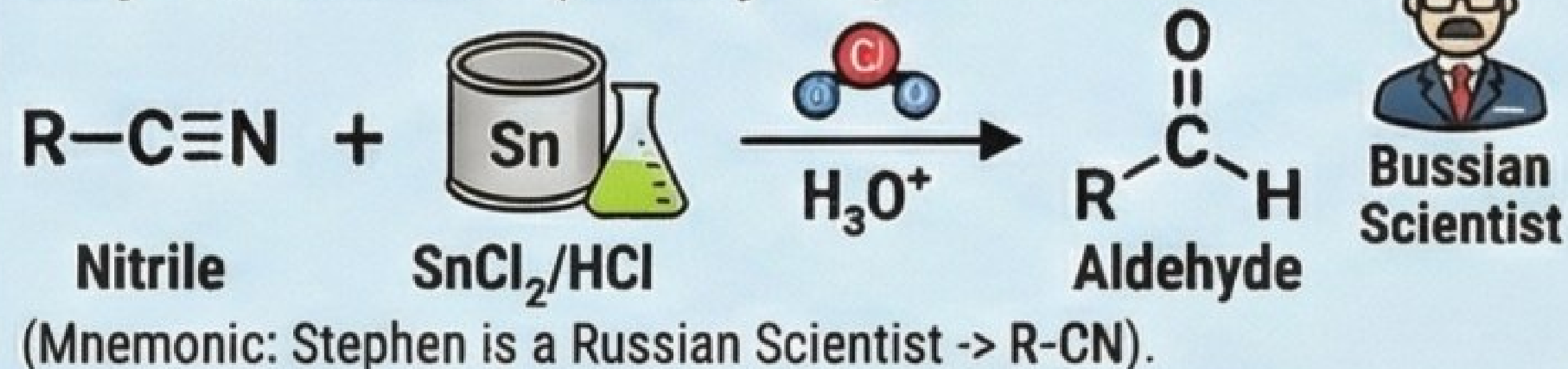


HOW TO PREPARE ALDEHYDES & KETONES

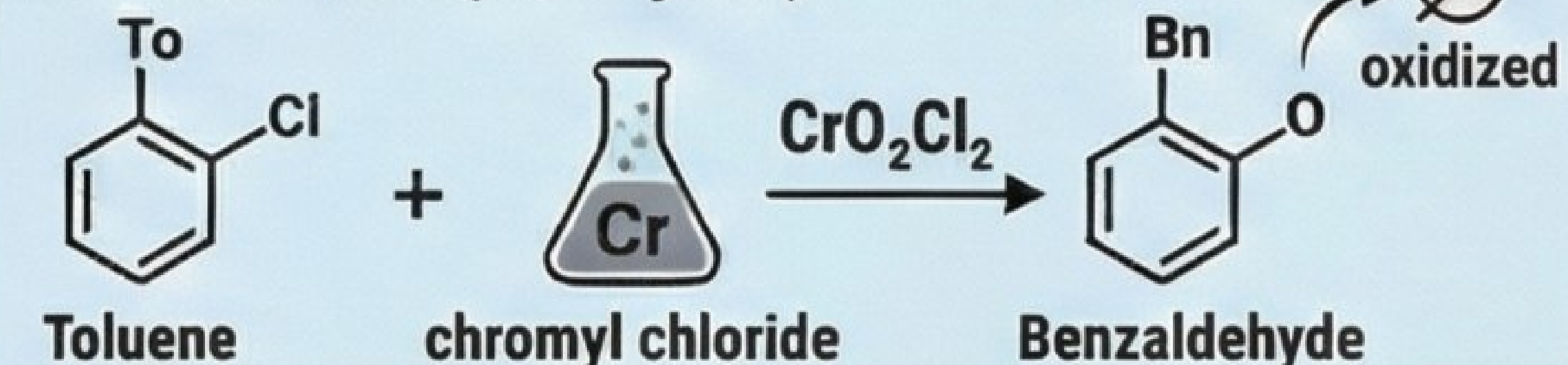
Rosenmund Reaction (Aldehydes)



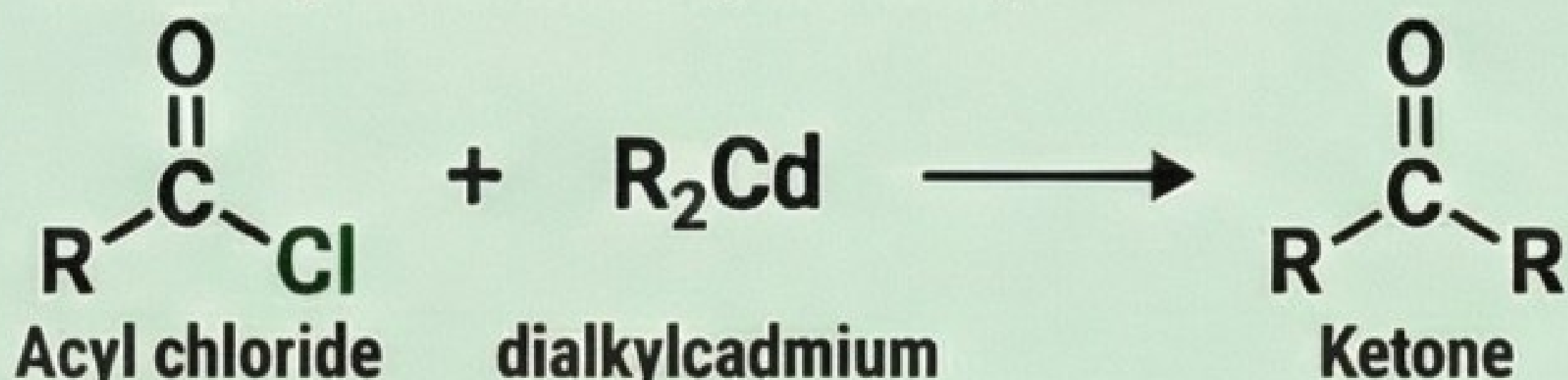
Stephen Reaction (Aldehydes)



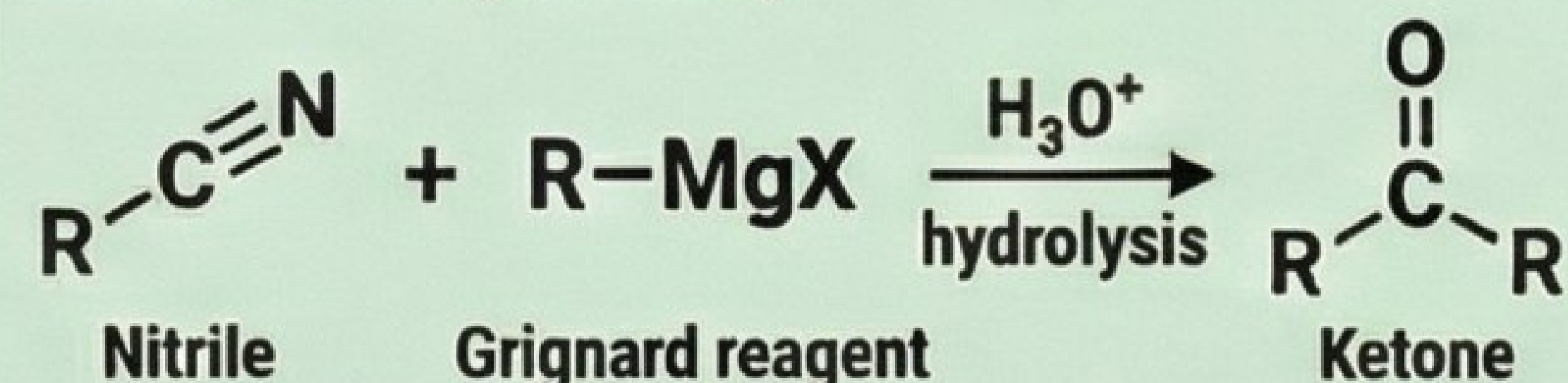
Etard Reaction (Aldehydes)



From Acyl Chlorides (Ketones)

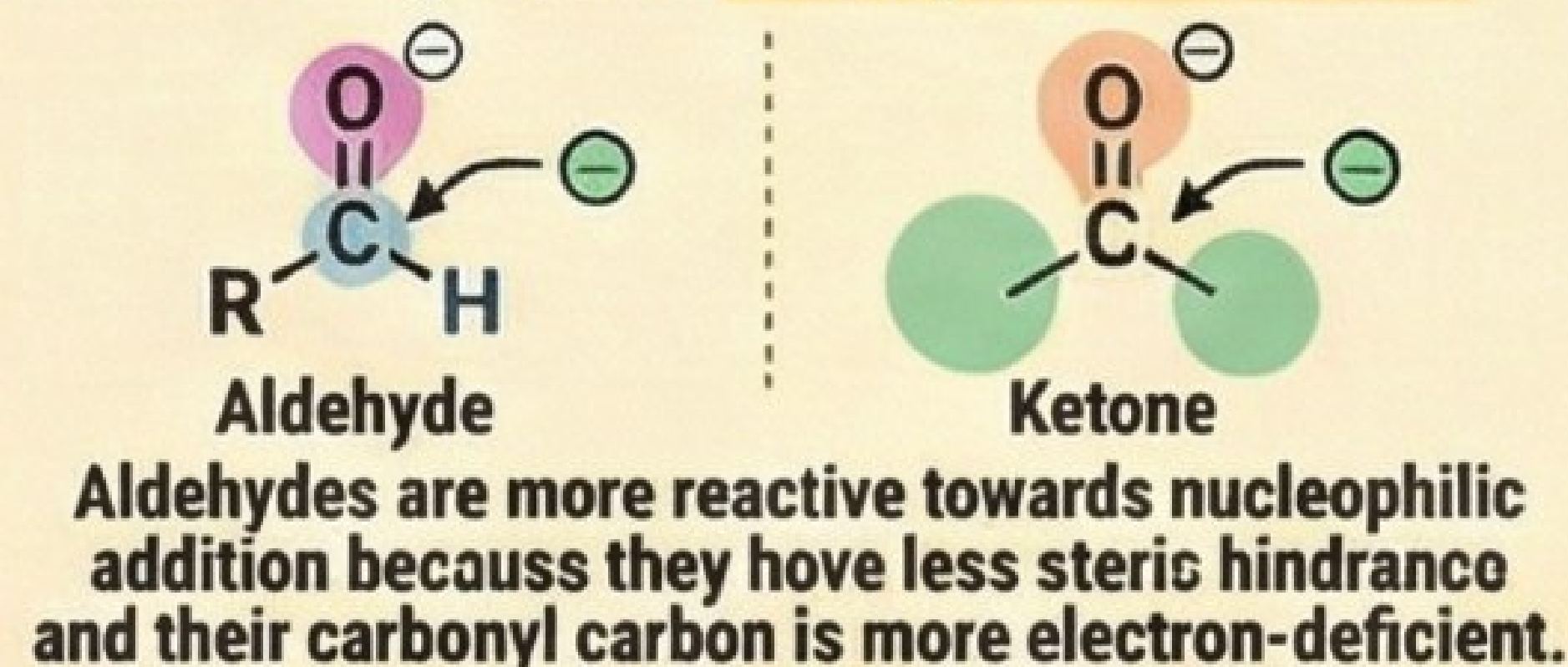


From Nitriles (Ketones)

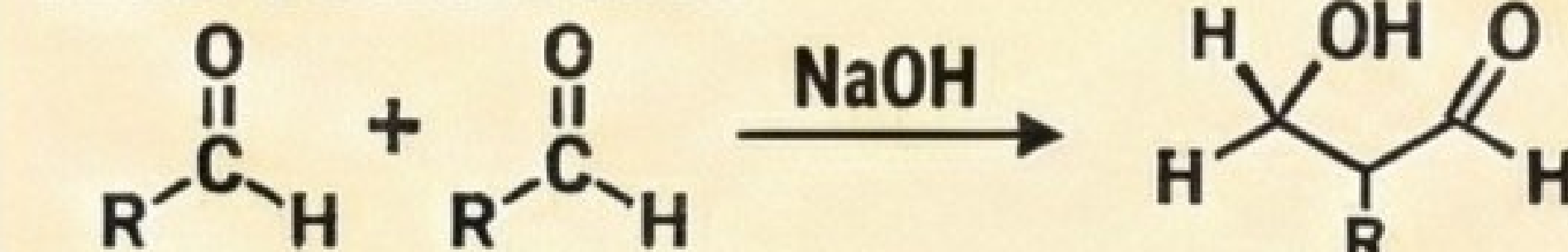


KEY REACTIONS OF ALDEHYDES & KETONES

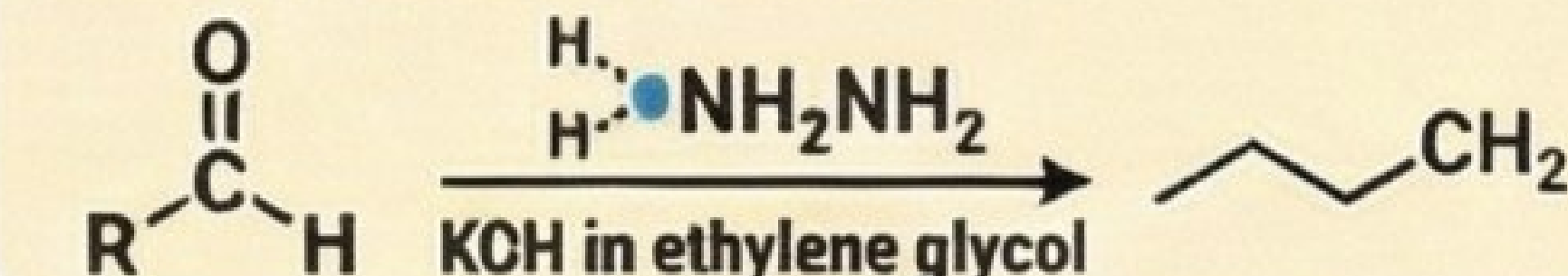
Reactivity Order: Aldehydes > Ketones



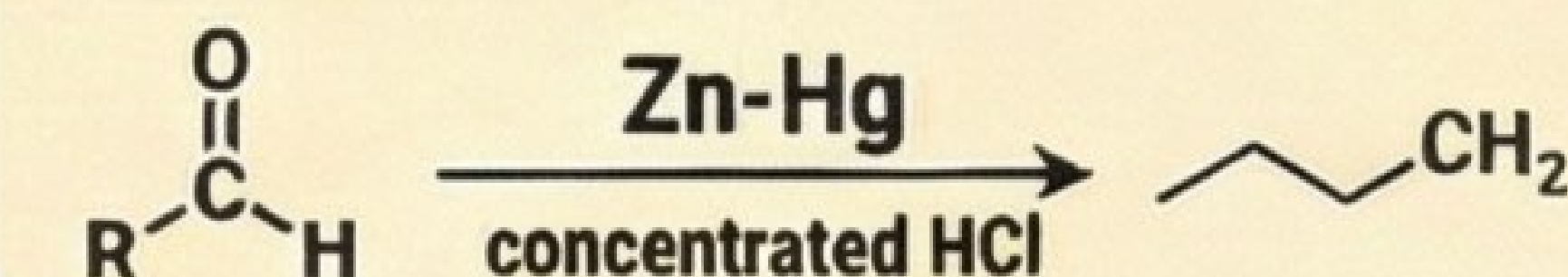
Aldol Condensation



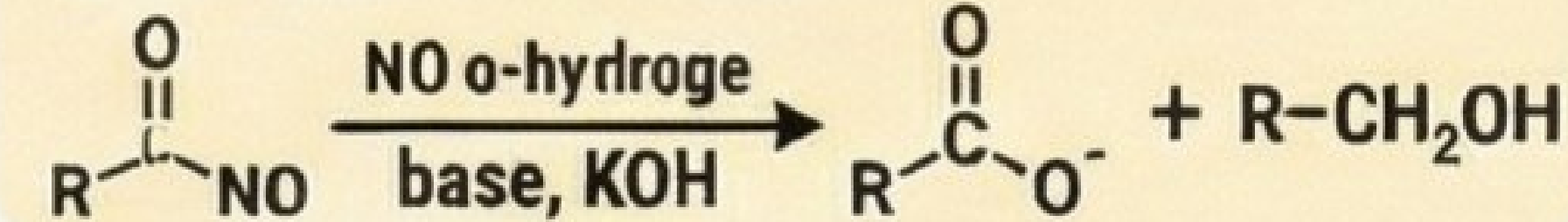
Wolf-Kishner Reduction



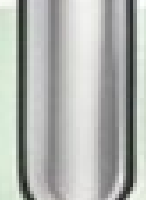
Clemmensen Reduction



Cannizzaro Reaction

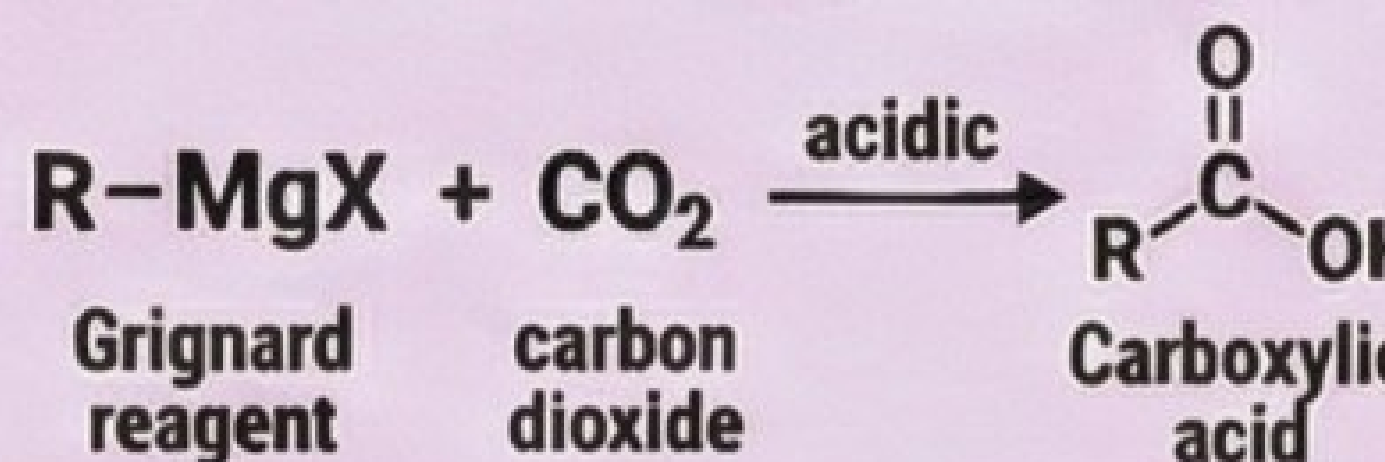


KEY DISTINGUISHING TESTS FOR ALDEHYDES

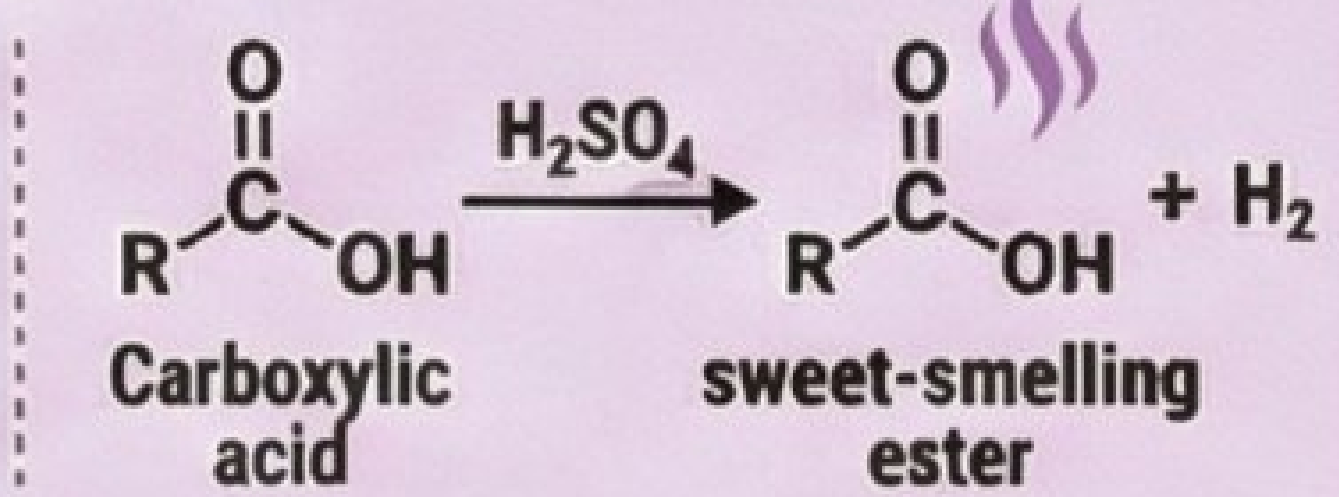
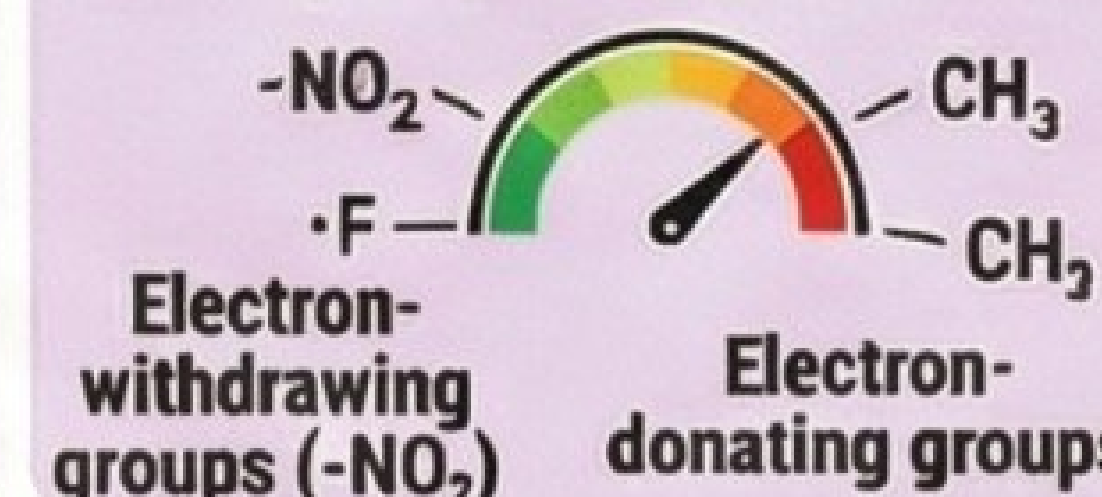
'Tollens' Test	Reagent: Silver Nitrate	Ammoniacal Silver Nitrate		Silver Mirror Forms a "Silver Mirror"	Yes
Fehling's Test	Reagent: Solution A + B	Fehling's Solution A + B		Red-brown Precipitate (Cu ₂ O)	No
Iodoform Test	Reagent: I ₂ / NaOH	Yellow Precipitate		Yellow Precipitate (CHI ₃)	Only for compounds with a CH ₃ -CO group

ALL ABOUT CARBOXYLIC ACIDS

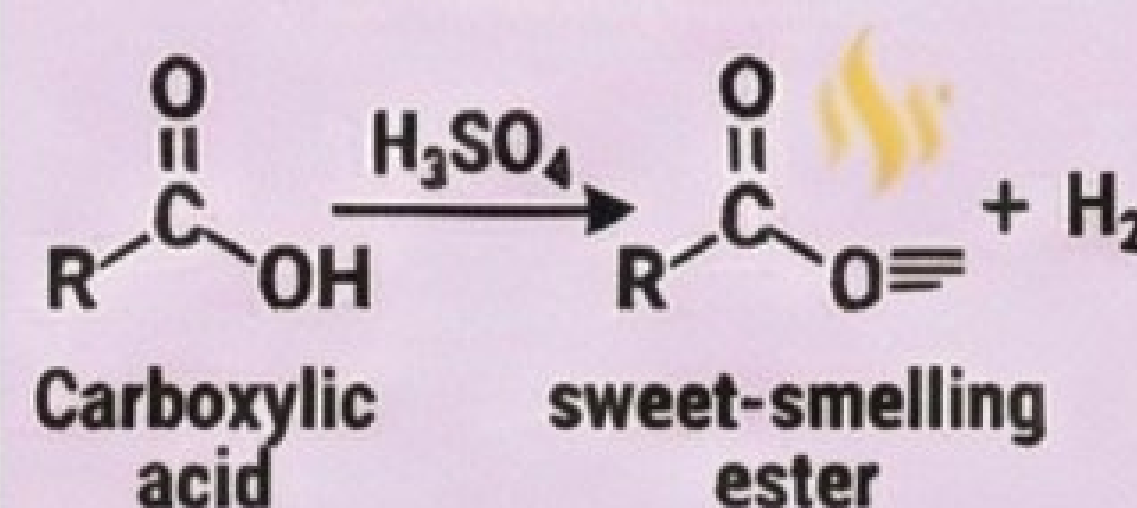
Preparation from Grignard Reagent



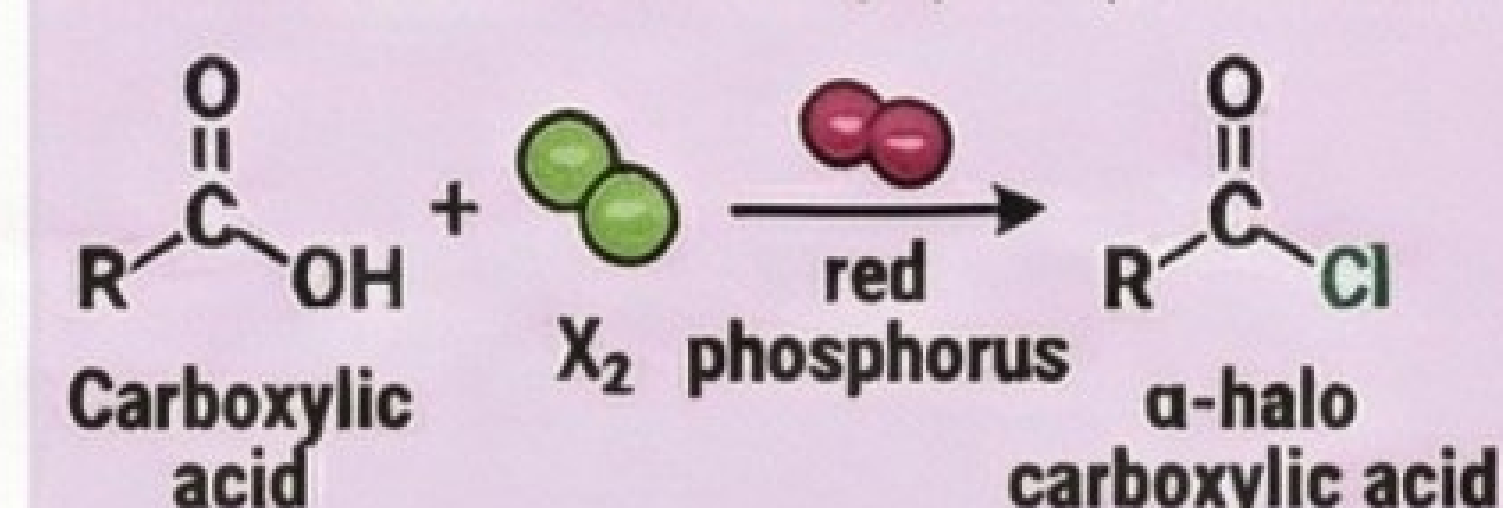
Acidity depends on attached groups



Esterification



Hell-Volhard-Zelinsky (HVZ) Reaction



Distinguishing from Phenol

